Clinical Risk Assessment Report: Patient 47

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 39.1% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.77x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +5.7%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.33 [Bottom 19%] (+5.7%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 1.24 [Top 8%] (+4.4%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.71 [Avg (70th %ile)] (+2.8%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.47 [Top 5%] (+2.4%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.21 [Bottom 16%] (+2.1%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: 1.83 [Top 7%] (+1.7%)**

Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: -1.14 [Bottom 13%] (+0.7%)**

Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET) **Value: 0.66 [Avg (75th %ile)] (+0.7%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)